

Higher Tier only questions

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Samples A, B and C contain magnesium or calcium ions ☒ Only samples A and B contain magnesium or calcium ions ☐ Only samples B and C contain magnesium or calcium ions ☐ None of the samples contain magnesium or calcium ions ☐			1	1		
	(b)			To show the expected result for permanent hard water ☐ To show the expected result for temporary hard water ☐ To act as a control for the experiment ☒ To make sure the investigation is a fair test ☐	1			1		1
	(c)			volume of soap solution and number of times shaken ☐ volume of water sample and volume of soap solution used ☐ volume of water sample and number of times shaken ☒ volume of water sample, volume of soap solution and number of times shaken ☐	1			1		1
	(d)			Sample B contains temporary hardness only ☐ Sample B contains permanent hardness only ☐ Sample B contains a mixture of temporary and permanent hardness ☒ Sample B does not contain hardness ☐ the volume of soap needed after boiling is less than before / less soap is needed after boiling (1) the volume of soap after boiling is still more than the distilled water (1)			3	3		3

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	(e)			if answered yes accept any two of following for (1) each • strengthens bones / teeth • reduced chance of heart disease • better for baking / brewing industry • improves flavour of water if answered no accept any two of following for (1) each • forms a scum with soap / wastes soap • furs up kettles / pipes / forms limescale • appliances become less efficient (or named appliance) • bad taste / tastes different if answered 'unable to decide' must have one reason in favour and one against to gain (2) if no opinion is stated, credit may still be awarded if the reasons clearly imply an opinion neutral: reference to cost / calcium / magnesium accept other creditworthy answers	2			2		
				Question 3 total	4	0	4	8	0	5